Real Time Gait Event Detection for Active Lower Limb Prosthesis for Different Terrains

Submitted by
Mahesh Satish Saboo
Soft Computing Laboratory
IIT Kharagpur
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Under the guidance of
Prof. Dilip Kumar Pratihar
IIT Kharagpur
Mentor: Mr. Saikat Sahoo

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Mahesh Saboo
Soft Computing Laboratory, IIT Kharagpur

Abstract:

Human gait phase detection plays an essential role in controlling the behavior of powered prosthesis for the lower limb. Depending upon the gait phase, the prosthesis can adjust its controlling parameters to provide a smooth transition for the amputee. This report presents a rule-based methodology to detect various gait events in real time. The method uses inputs from two IMUs attached to the shank and ankle of the subject. The method can recognize nine gait events for level ground walking, namely – Early Heel Strike, Heel Strike, Early Heel Max, Heel Max, Early Heel Off, Heel Off, Early Toe Off, Toe Off and Early Mid Swing, with a detection accuracy of 100 percent. Next, the report presents a method to detect six gait events for stair ascend and descend motion with a detection accuracy of 100 percent.

Keywords: Gait Phase | Prosthesis | Rehabilitation Robotics

1. Nomenclature

Abbreviation	Meaning
LGW	Level Ground Walking
EHS	Early Heel Strike
HS	Heel Strike
EHM	Early Heel Max
HM	Heel Max
EHO	Early Heel Off
HO	Heel Off
ETO	Early Toe Off
TO	Toe Off
EMS/MS	Early Mid Swing
ST/TS	Strike
TM	Toe Max
ETM	Early Toe Max
ES/EES	Early Strike
AAVY	Ankle Angular Velocity (Y)
SAVZ	Shank Angular Velocity (Z)
SANX	Shank Angle (X)
SACX	Shank Acceleration (X)

2. Introduction

Along with the design of powered ankle prosthesis, a suitable control algorithm is also needed for the movement of the motorized ankle joint. The modern approach to control the powered lower limb prostheses is the supervisory control, where the intent of the user is recognized in order to design a robust and precise control strategy. This type of supervisory control works in a hierarchy of three stages: the first stage is the supervisory control, the second stage is considered as finite state control and low-level control or the torque control is considered as the third stage. The powered prosthesis is able to mimic the behavior of an intact ankle joint with a suitable coordination among these three stages [1]. Most of the existing gait event detection algorithms based on offline detection. Detecting gait events in real time is different from detecting gait events in an offline mode such as [2] as real-time needs high-speed computation and hence a simpler algorithm. [3] uses a rule-based approach to detect some of the gait events in real time. Most of the studies in the field have been for level ground or inclined slope. The method developed is novel in the way that it uses a simple rule-based approach to detect gait events for stair ascend and descend motions along with level ground walking. The first section provides the details of the algorithm developed for level ground walking. Next, the report present algorithms for gait event detection for stair ascend and descend motions.

3. Level Ground Walking

The walking mechanism of an able-bodied human repeats the events shown in Figure 1 in a cyclic manner and this cycle is termed as gait cycle. From the perspective of the prosthesis, Heel Strike, Heel Max, Heel Off, Toe Off and Mid Swing plays an important role in the control of the prosthesis.

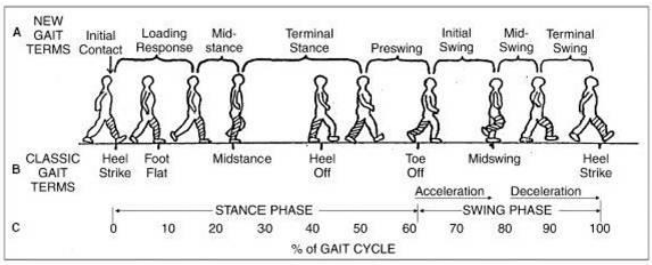

Figure 1 Human gait cycle

3.1. Problem Statement

The objective of the experiments performed for LGW involves the detection of nine gait events. Following are the events to be detected:

- EHS: This event marks the period after actual MS and before actual HS. It provides the prosthesis information about an upcoming heel strike.
- HS: This event marks the instance when the heel of the subject makes a contact with the ground. It provides the prosthesis information about an actual heel strike.
- EHM: This event marks the period after actual HS and before actual HM. It provides the prosthesis prior information for high force on heel.

- HM: This event marks the instance when the force on the heel of the subject is maximum in the period between HS and HO.
- EHO: It provides the prosthesis prior information for the heel off event.
- HO: This event marks the instance when the heel of the subject is off the ground.
- ETO: It provides the prosthesis prior information for the toe off event.
- TO: This event marks the instance when the toe of the subject is off the ground.
- EMS: It provides the prosthesis a prior information for the mid swing event. This information also marks the end of a complete gait cycle that began with EHS event.

3.2. Experimental Protocol

A wearable was designed for collecting the data for the experiment. It included an Arduino-Nano micro-controller, an onboard Raspberry Pi 3 computer, an I2C multiplexer, 2 Force Sensitive Resistors (FSRs) placed at the heel and the toe of the subject and 2 Inertial Measurement Units (IMUs) attached to the shank and the ankle of the subject. The data was collected at the frequency of 40 Hz. 10 Subjects were asked to walk for over 100 steps at their normal walking speeds with each experiment lasting for not more than 40 steps. At the beginning of each data collection experiment, the IMUs were calibrated for a straight standing position for 20 seconds duration. This average value was then subtracted from the observed value in real time to get the calibrated value as the IMUs used were of absolute orientation. The primary axis for ankle IMU is the Y-axis and for shank IMU is the X-axis. The experimental setup can be seen in Figure 2.

3.3. Extraction of Actual Features

The actual gait events are detected using the FSRs placed on the underside of the foot. As the force on FSR changes, the resistance of the FSR changes and thus the voltage output. This change in voltage helps us to detect the actual events according to the following rules:

- Heel Strike: When three consecutive non zero FSR (Heel) readings are observed, we mark the first reading as HS. We take three consecutive terms so as to ignore the unwanted noise in the sensor reading.
- Heel Max: This is the maximum of the FSR (Heel) readings between the HS and HO.
- Heel Off: When we observe a zero FSR (Heel) reading after HS, we consider it to be the instance of HO.
- Toe Off: When we observe a zero FSR (Toe) reading after HO, we consider it to be the instance of HO.
- Mid Swing: It is marked by the sample exactly in the middle of a TO and consecutive HS.

3.4. The Algorithm

A rule based algorithm was developed to detect the gait phases. The algorithm consists of 11 checkpoints, each marking a feature of human walking pattern. These checkpoints are then run in a loop to detect phases in numerous gait cycles. Table 1 shows the details of each checkpoint.

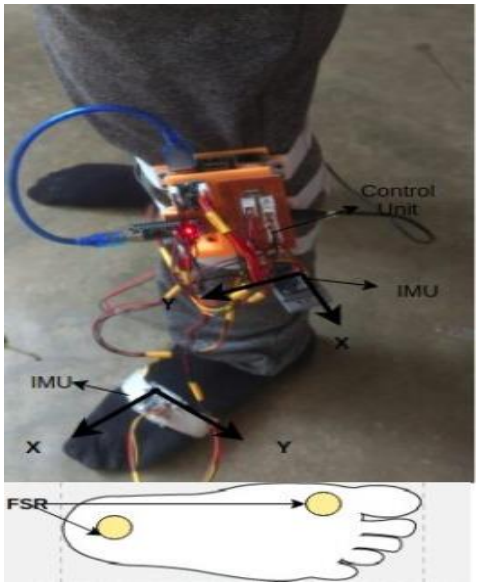

Figure 2 The experimental setup

CP	Condition	Remark
CP1	• $(CP1 < CP11)$ AND $(SAVZ < -50)$ AND (SAVZ is increasing)	
CP2	• $(CP2 < CP1)$ AND $(SAVZ > -100)$	This marks EHS
CP3	• $(CP3 < CP2)$ AND $(AAVY > 0)$	
CP4	• $(CP4 < CP3)$ AND $(abs(SACX) > 1)$	This marks HS. Angle = SANX
CP5	• $(CP5 < CP4)$ AND $(AAVY < 50)$ AND (AAVY is decreasing)	This marks EHM
CP6	• $(CP6 < CP5)$ AND (3 Consecutive $abs(AAVY) < 10$)	This marks HM
CP7	• $(CP7 < CP6)$ AND (SAVZ is increasing)	This marks EHO
CP8	• $(CP8 < CP7)$ AND • (2 Consecutive $AAVY > 1$)AND (SAVZ is increasing) AND $(SANX < 0)$ OR $(SANX < Angle - 25)$	This marks HO
CP9	• $(CP9 < CP8)$ AND $(AAVY > 50)$ AND (AAVY is increasing)	This marks ETO
CP10	• $(CP10 < CP9)$ AND • $(AAVY > 200)$ OR $(SANX < Angle - 50)$OR $(AAVY > 10)$ AND (AAVY is decreasing) AND (SAVZ is increasing)	This marks TO
CP11	• $(CP11 < CP10)$ AND $(SAVZ < 0)$	This marks EMS

Table 1 Rule base for event detection for LGW

3.5 Results

Here, as the frequency is 40 Hz, the sampling frequency in time domain can be considered to be 25 milliseconds. The algorithm has been tested for 1072 steps over 10 different able-bodied subjects. Table 2 shows the average error in the detection of the gait events. The results show that the detection accuracy is 100 percent with close to 1 sample for HS, close to 2 samples for HM and HO, and around 1.5 samples for TO. As the detection accuracy is 100 percent, we can use any of these phases to detect the step frequency and thus this algorithm can also give us step count for the person. The algorithm uses 4 inputs from the 2 IMUs namely: AAVY, SAVZ, SANX and SACX.

Event	Detection Accuracy	Average Detection Error (in ms)	Average Detection Error (in samples)	Average Early Detection (in ms)
HS	100 %	27	1.08	190
HM	100 %	50	2	110
HO	100 %	52	2.08	275
TO	100 %	36	1.44	118
EMS	100 %	----	----	114

Table 2 Results for gait event detection for LGW

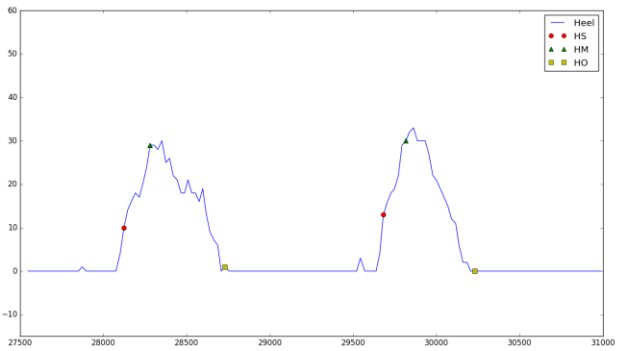

Figure 3 Event detection for actual heel events for LGW

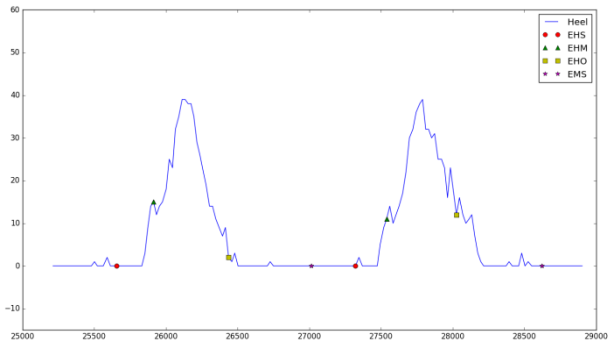

Figure 4 Event detection for early heel events for LGW

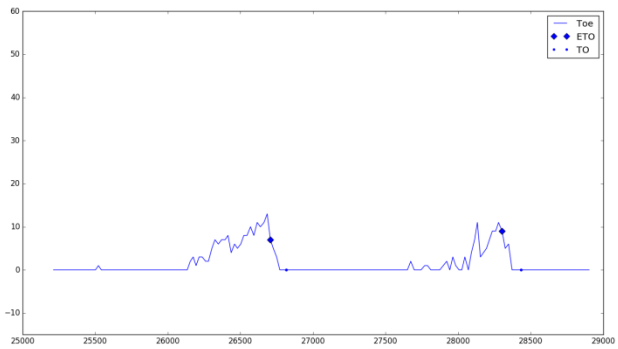

Figure 5 Event detection for toe events for LGW

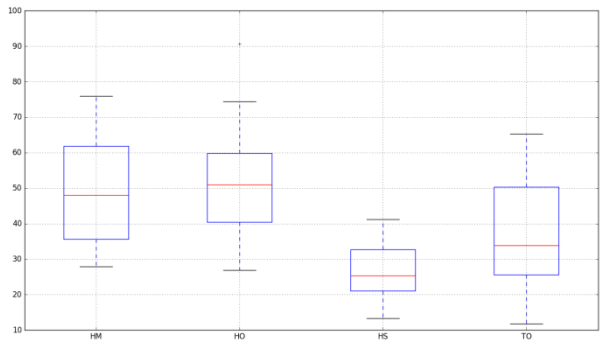

Figure 6 Box plot description of detected events for LGW

The graphs in Figures 3, 4 and 5 shows the detection of events for gait cycles. Figure 3, shows detection of HS, HM, and HO using the above algorithm. HM and HO almost coincide with the actual HM and HO but the figure misleads in terms of HS as at the instance of HS, there will be some force on the Heel HSR and thus the point is placed above zero line. Figure 4 confirms that our algorithm predicts the HS, HM and HO fairly early so as to decide upon the control action of the prosthesis. Figure 5 confirms our algorithm working for toe off event. Figure 6 shows a plot describing the variations in the detection of various events. The variation for accuracy of detection is found to be 27 ± 7.22 ms for HS, 50 ± 15.2 ms for HM, 52 ± 15 ms for HO and 36 ± 14.6 ms for TO. Please note, here average is calculated as absolute difference from actual and hence all values are positive. The accuracy can also be described in terms of RMSE errors by using actual difference instead of absolute.

4. Stair Ascend and Descend

The walking mechanism for stairs differs from that for LGW. In LGW, the Heel strikes the ground before the toe, while for the motion on stairs either of the toe or the heel can strike the ground and thus we term this as Strike and not Heel Strike or Toe Strike. For motion on stairs, the force on the toe is higher than that on the heel and hence, here we detect the toe max and toe-off events. In most of the cases, we observe two peaks in the toe force. First one is observed just after the strike and is not always a prominent peak. The second one is observed just before toe-off and is always a prominent peak with a large force acting on the toe. Here, we refer to this second prominent peak as toe max. You can see the two peaks in Figure 7.

4.1 Problem Statement

The objective of the experiments performed for stairs involves the detection of the six gait events. Following are the detected events:

- EST: This event marks the period after the actual MS and the before actual ST. It provides the prosthesis information about an upcoming strike.
- ST: This event marks the instance when the heel or the toe of the subject makes contact with the ground. It provides the prosthesis information about an actual strike.
- ETM: This event marks the period after actual ST and before actual TM. It provides the prosthesis prior information for high force on toe.
- TM: This event marks the instance when the force on the toe of the subject is maximum in the period just before TO.
- TO: This event marks the instance when the toe of the subject is off the ground.
- EMS: It provides the prosthesis prior information for the mid swing event. This information also marks the end of a complete gait cycle that began with EST event.

4.2. Experimental Protocol

The data was collected at the frequency of 40 Hz using the same wearable that was used in the case of LGW. 10 Subjects were asked to ascend and descend the stairs multiple times at their normal speeds with each experiment lasting for not more than 7 steps. At the beginning of each data collection experiment, the IMUs were calibrated for a straight standing position for 20 seconds duration as the IMUs used were of absolute orientation.

4.3. Extraction of Actual Features

The actual gait events are detected using the FSRs placed on the underside of the foot. As the force on FSR changes, the resistance of the FSR changes and thus the voltage output. This change in voltage helps us to detect the actual events according to the following rules:

- Strike: This event is observed when we observe four consecutive non zero FSR (Toe or Heel or Both) values.
- Toe Max: Maximum of 10 samples of Toe FSR before the TO event.
- Toe Off: It is marked as three consecutive zero FSR Toe readings after AAVY > 100 is found post ST.
- Mid Swing: This event is observed at the exact middle sample of a TO and consecutive ST.

Note: Due to sensor sensitivity issues, we have to incorporate the IMU data to find the actual instances of the events. For more details, refer the codes and graphs.

4.4. The Algorithm

A rule-based algorithm was developed to detect the gait events. The algorithm consists of multiple checkpoints, each marking a feature of human walking pattern. These are then run in a loop to detect events in numerous gait cycles. The conditions for detection of the events for stair ascend and descend are tabulated in the Table 3 and 4 mentioned below.

CP	Condition	Remark
CP1	• $(\mathrm{CP1} < \mathrm{CP10})$ AND $(\mathrm{SAVZ} < -100)$ AND (SAVZ is increasing)	This marks EST
CP2	• $(\mathrm{CP2} < \mathrm{CP1})$ AND • $(\mathrm{SAVZ} > 0)$ OR $(\mathrm{abs}(\mathrm{SACX}) > 6)$	
CP3	• $(\mathrm{CP3} < \mathrm{CP2})$ AND $(\mathrm{abs}(\mathrm{SACX}) < 1)$	This marks ST
CP4	• $(\mathrm{CP4} < \mathrm{CP3})$ AND (a negative minimum of SAVZ is found)	
CP5	• $(\mathrm{CP5} < \mathrm{CP4})$ AND $(\mathrm{SAVZ} > 0)$	This marks ETM
CP6	• $(\mathrm{CP6} < \mathrm{CP3})$ AND $(5 \text{ times } \mathrm{abs}(\mathrm{AAVY}) < 5)$	
CP7	• $(\mathrm{CP7} < \mathrm{CP6})$ AND $(\mathrm{AAVY} > 50)$ AND (AAVY is increasing)	This marks TM
CP8	• $(\mathrm{CP8} < \mathrm{CP7})$ AND • $(\mathrm{AAVY} > 220)$ OR (AAVY is decreasing)	This marks TO
CP9	• $(\mathrm{CP9} < \mathrm{CP8})$ AND $(\mathrm{SAVZ} > 50)$	
CP10	• $(\mathrm{CP10} < \mathrm{CP9})$ AND $(\mathrm{SAVZ} < 0)$	This marks EMS

Table 3 Rule base for detection of events for stair ascend motion

CP	Condition	Remark
CP1	• $(CP1 < CP8)$ AND $(SAVZ < -100)$ AND (SAVZ is increasing)	
CP2	• $(CP2 < CP1)$ AND $(SAVZ > 0)$	This marks EST Angle = SANX
CP3	• $(CP3 < CP2)$ AND $(SANX < Angle - 20)$	This marks ETM
CP4	• $(CP4 < CP2)$ AND (3 times abs(AAVY) < 10)	
CP5	• $(CP5 < CP4)$ AND $(AAVY > 60)$ AND (AAVY is increasing)	This marks TM
CP6	• $(CP6 < CP5)$ AND • $(AAVY > 220)$ OR (AAVY is decreasing)	This marks TO
CP7	• $(CP7 < CP6)$ AND $(SAVZ > 50)$ AND (SAVZ is decreasing)	
CP8	• $(CP8 < CP7)$ AND $(SAVZ < -100)$	This marks EMS

Table 4 Rule base for detection of events for stair descend motion

Event	Detection Accuracy	Average Detection Error (in ms)	Average Detection Error (in samples)	Average Early Detection (in ms)
ST	100 %	83	3.3	295
TM	100 %	27.4	1.1	256
TO	100 %	33.4	1.3	NA
EMS	100 %	NA	NA	64

Table 5 Results for event detection for stair ascend motion

Event	Detection Accuracy	Average Detection Error (in ms)	Average Detection Error (in samples)	Average Early Detection (in ms)
ST	100 %	NA	NA	197
TM	100 %	55.5	2.2	275
TO	100 %	49.4	1.98	NA
EMS	100 %	NA	NA	210

Table 6 Results for event detection for stair descend motion

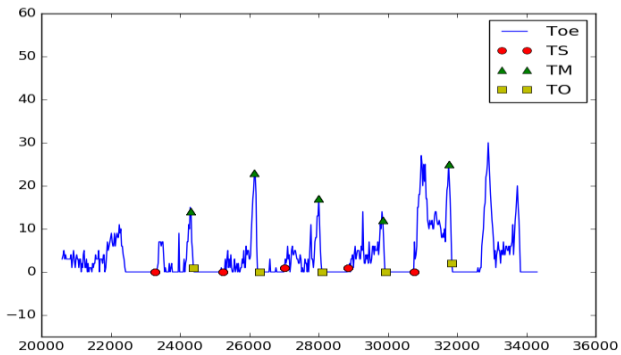

Figure 7 Event detection for actual events for stair ascend

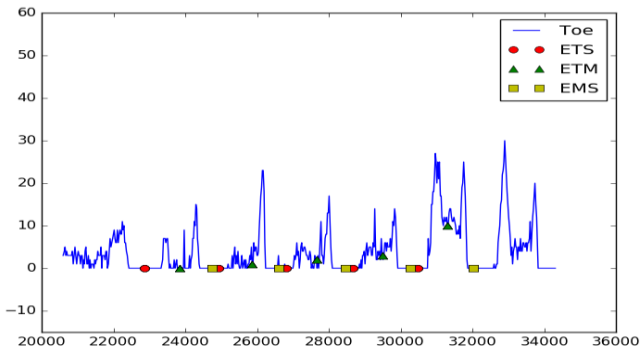

Figure 8 Event detection for early events for stair ascend

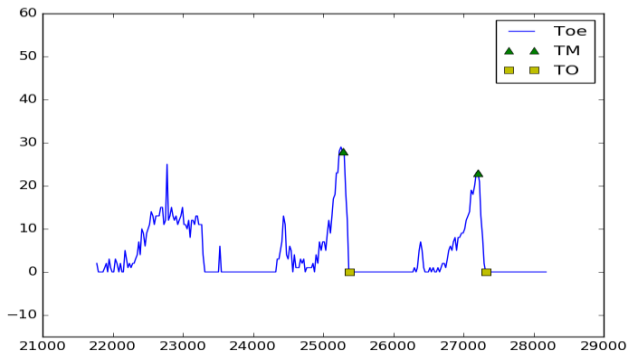

Figure 9 Event detection for actual events for stair descend

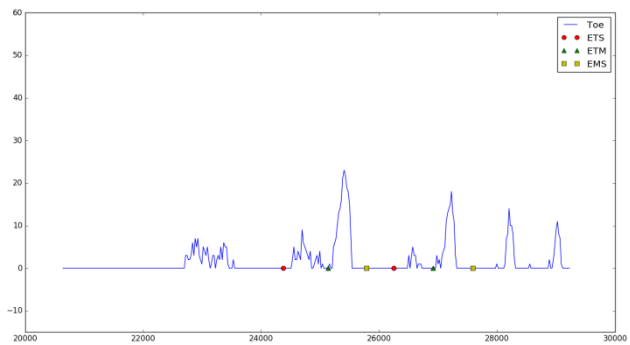

Figure 10 Event detection for early events for stair descend

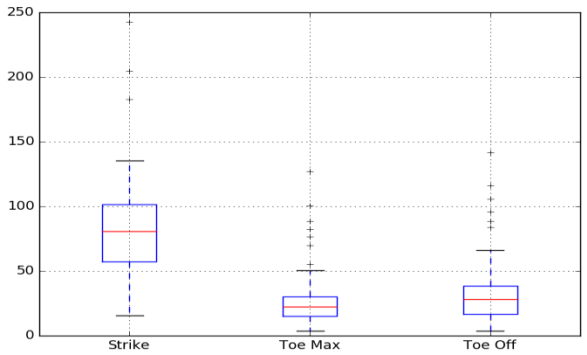

Figure 11 Boxplot description of detected events for stair ascend

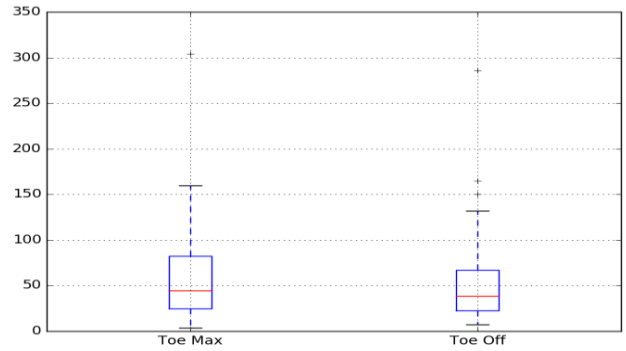

Figure 12 Boxplot description of detected event for stair descend

4.5 Results

Here, as the frequency is 40 Hz, the sampling frequency in the time domain can be considered to be 25 milliseconds. The algorithm has been tested for 529 steps of over 10 different able-bodied subjects for stair ascend motion and 506 steps of over 10 different subjects for stair descend motion. Table 5 and 6 show the average error in the detection of gait events. The results show that the detection accuracy is 100 percent with close 3.5 samples for ST, 1.1 samples for TM and around 1.3 samples for TO in the case of stair ascend motion. Also, in the case of stair descend, we were able to detect TM and TO with higher accuracy of close to 2 samples but failed to detect ST with such high accuracy. As the detection accuracy is 100 percent, we can use any of these events to detect the step frequency and thus this algorithm can also give us step count for the person. The algorithm uses 4 inputs from the 2 IMUs namely: AAVY, SAVZ, SANX and SACX. Figure 7 and 9, shows detection of ST, TM, and TO for the cases of stair ascend and descend using the above algorithm. Figure 8 and 10 confirms that our algorithm predicts the EST, ETM and EMS fairly early so as to decide control action of the prosthesis in a particular phase.

Figure 11 and 12 describes the variations in the detection of various events. The variations for accuracy of detection is found to be 83 ± 35.8 ms for ST, 27.5 ± 20.5 ms for TM and 33.4 ± 24.5 ms for TO in the case of stair ascend whereas in the case of stair descend the variations are 55.6 ± 44.7 ms for TM and 49.4 ± 40.4 ms for TO. Please note, here average is calculated as the absolute difference from actual and hence all values are positive. We can also describe the accuracy in terms of RMSE errors.

5. Conclusion

The aim of this study was to develop and assess an IMU based algorithm for real-time gait event detection on different terrains. To assess the performance of the newly proposed algorithm in detecting gait events during walking, the conventional FSR based foot switches were used as a reference method. As an alternative to the FSR sensor for gait event detection, an IMU could provide reliable gait event signals, and it requires relatively low power consumption, is low cost, and is more convenient to wear. The proposed IMU based algorithm could detect gait events with robustness and accuracy in real-time applications as proved from the above results. To the best of our knowledge, the gait event detection on stair terrains (ascending and descending stairs) has rarely been investigated such as [4]. Hence, our proposed algorithm may have the potential for accurate gait event detection on stair terrains in real-time applications in comparison to the previous methods.

This report proposed a new algorithm to detect gait events on three walking terrains in real-time based on the analysis of inputs from a 9 DOF IMU. The accuracy and robustness of the algorithm were validated by data from ten healthy subjects while walking on level ground, upstairs, and downs stairs terrain. The experimental results showed that the proposed algorithm can accurately detect gait events with comparable accuracy and time delays across different terrains and thus can be used in prosthesis purpose.

Acknowledgement

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